

43-IAE : 43rd Indian Scientific Expedition to Antarctica

Region: Antarctica | Window: 114 Days (Nov 2023 - Mar 2024) | Status: Under Analysis

Lead Scientist:	Dr. Vikramaditya Sen (NCPOR)
Platform / Base:	Chartered Ice-Class Vessel MV Vasilii Golovnin
Team Size:	48 Field Researchers & Technicians
Research Focus:	Glaciology, Atmospheric Physics, Geodesy, Cryospheric Geobiology

1. MISSION OVERVIEW & SUMMARY

Conducted comprehensive glaciological traverse across the Polar Ice Sheet, deployed automated weather buoys, drilled 120m ice cores near Dronning Maud Land, and replenished Bharati and Maitri research infrastructure. Operations traversed 1,800 km on snow cats through extreme sub-zero weather.

2. PRIMARY SCIENTIFIC OBJECTIVES

- [1] Extract 120m deep ice cores for paleoclimate reconstruction spanning 2,000 years
- [2] Deploy autonomous GPS receivers for continuous ice-sheet flow tracking
- [3] Conduct aerosol and cloud condensation nuclei sampling over Princess Elizabeth Land
- [4] Maintain year-round atmospheric and geomagnetic recording instruments